MOD_A MODULO TORSION MODULES IS CANONICALLY EQUIVALENT TO Q(A)-MODULES, Q(A) THE TOTAL QUOTIENT RING

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The proof is a two-line specialization of proposition-localization-and-serre-quotients (delivered card L48282 of this lane); the nontrivial identification 'torsion iff every element annihilated by a non-zerodivisor' is stated in the proof. Content verbatim.

Editorial note. The label proposition-quotient-by-torsion-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- Serre subcategories and quotient categories; the localization equivalence Mod_A/T -> Mod_{S^{-1}A} for a Serre subcategory of S-torsion modules (same-chapter proposition-localization-and-serre-quotients, delivered as L48282)
- The total quotient ring Q(A) = S^{-1}A with S the non-zerodivisors (Algebra, example-localize-at-prime); torsion module = every element annihilated by some non-zerodivisor

Source reference. Original section 02C5: sources/stacks-chow/tags.txt, source lines 3072–3072. Prerequisite pointer invoked at 6820; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EA5: sources/stacks-examples/pinned-original-source.tex, source lines 6784–6784. Prerequisite pointer invoked at 6830; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let A be a ring. Let $Q(A)$ denote its total quotient ring (as in Algebra, Example 02C5). Let Mod_A denote the category of A-modules and $\mathcal{T}$ its Serre subcategory of torsion modules. Let $Mod_{Q(A)}$ denote the category of $Q(A)$-modules. Then there is a canonical equivalence $Mod_A/\mathcal{T} \to Mod_{Q(A)}$.*

Proof. Follows immediately from applying Proposition 0EA5 to the multiplicative subset $S = \{f \in A \mid f \text{ is not a zerodivisor in } A\}$, since a module is a torsion module if and only if all of its elements are each annihilated by some element of S. □

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